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Hybrid grid/basis set discretizations of the Schrödinger equation

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We present a new kind of basis function for discretizing the Schrödinger equation in electronic structure calculations, called a gausslet, which has wavelet-like features but is composed of a sum of Gaussians. Gausslets are placed on a grid and combine advantages of both grid and basis set approaches. They are orthogonal, infinitely smooth, symmetric, polynomially complete, and with a high degree of locality. Because they are formed from Gaussians, they are easily combined with traditional atom-centered Gaussian bases. We also introduce diagonal approximations that dramatically reduce the computational scaling of two-electron Coulomb terms in the Hamiltonian. *Published by AIP Publishing.* https://doi.org/10.1063/1.5007066

I. INTRODUCTION

Most numerical simulations of the Schrödinger cannot work directly in the continuum; they require a discretization of some sort that maps the continuum to a finite number of degrees of freedom. For example, in the many thousands of electronic structure calculations performed annually for molecules and solids, basis sets are usually used. For molecules, these are usually atom centered functions composed of linear combinations of Gaussian type functions, for which analytic integrals are used to quickly generate the discrete Hamiltonian terms. For solids, plane wave basis sets are often used, which also have analytic integrals. These basis set methods have been developed and refined over most of a century, and the software to use them has been improved to a remarkable degree. Nevertheless, a variety of alternative discretization methods such as wavelet bases,[1–3] adapted grids,[4,5] finite elements,[6] and sinc-function bases[7] have continued to be developed. These alternatives have been pursued to address real weaknesses in existing discretizations, but Gaussian and plane wave bases are still used for most calculations.

What is the motivation to try to make an improved discretization approach again? Our primary reason is that some of the most promising new strong-correlation methods for electronic structures—the density matrix renormalization group (DMRG)[8–10] and related tensor network methods[11]—depend strongly on the discretization. These methods have strong requirements for *locality* in the basis, due to the low-entanglement nature of their approximations. One can use localized Gaussian basis sets for DMRG, but this combination does not perform nearly as well as DMRG for simple lattice models, like the Hubbard model. The ideal basis for a DMRG or tensor network electronic structure calculation would be some sort of hypothetical real space grid with a very modest number of grid points, comparable to the number of functions in an atom-centered Gaussian basis, and giving comparable accuracy. Besides the DMRG, some other approaches in the electronic structure, such as linear scaling methods, also benefit or require high locality and grid or grid-like representations, and this has stimulated substantial work in this general area. However, the specific requirements of DMRG and related methods, which differ from other correlation methods, leave room for new approaches distinct from previous work.

Our approach has two key elements. The first is a novel type of basis function, called a *gausslet*, related to the scaling functions of wavelet approaches, designed to live on a uniform grid. Several features of these functions are particularly nice for electronic structures, particularly that they are orthogonal, localized, smooth, symmetric, and composed of a sum of Gaussians. The second element is a set of *diagonal* approximations for the Hamiltonian elements. These sort of diagonal approximations have previously been developed for sinc-function bases,[7] but we show that these approximations also arise in another context, in terms of wavelet-type bases with certain moment properties, such as our gausslets. The gausslets are much more localized than sinc functions, making them better suited for DMRG. In making a diagonal approximation for the two-electron interaction, the number of terms (integrals) needed drops from N^4 to N^2, where N is the number of basis functions. This reduction is not a large-N asymptotic feature—it appears for any N with no large coefficient in front of N^2. There are other recent approaches, for example, utilizing density fitting and tensor factorization,[12,13] that also produce $O(N^2)$ representations of the two-electron interaction. The DMRG might be adapted to use these representations; our approach would not require any changes to the standard DMRG. The gausslets and diagonal approximations combine to take us a substantial way towards the performance of the ideal very coarse but accurate real space grid.

Section II gives a more detailed overview of the problem and the approach. Section III discusses grids of Gaussians as bases, which have nearly ideal completeness properties but suffer from near linear dependence. Section IV discusses ternary wavelet transformations (WTs) that are used in Sec. V to derive gausslets. Section VI presents diagonal approximations, and Sec. VII has conclusions and a discussion of further directions. Throughout we give numerical examples in 1D.

II. OVERVIEW

Consider an arbitrary basis set approach. The Hamiltonian is an operator with the coefficients of the operator terms defined by integrals over the basis. By far, the most complex part of the Hamiltonian is the two-electron Coulomb interaction operator, parameterized by a four-dimensional array or tensor

$$V_{ijkl} = (ij|kl) = \int_{\mathbf{r}_1}\int_{\mathbf{r}_2} \frac{\phi_i(\mathbf{r}_1)\phi_l(\mathbf{r}_1)\phi_j(\mathbf{r}_2)\phi_k(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}, \quad (1)$$

where $\phi_i(\mathbf{r})$ make up the set of N basis functions, $i = 1, \ldots, N$. In some simpler approximations (e.g., the local density approximation of density functional theory), one can bypass the use of V_{ijkl} in favor of, say, solving the Poisson equation, but usually for more accurate methods treating electron-electron correlation explicitly, one cannot avoid dealing with V_{ijkl}, which has N^4 elements.

In contrast, a simple cubic grid discretization gives a representation of the electron-electron interaction which scales only as N^2, where now N is the number of grid points. In such an approach, one could use finite-difference approximations to represent kinetic energy derivatives. The nucleus-electron and electron-electron interactions would be evaluated pointwise, with the two-electron Hamiltonian terms taking the form $V_{ij}\hat{n}_i\hat{n}_j$, where $\hat{n}_i$ is the density operator on site i, and $V_{ij} = |\vec{r}_i - \vec{r}_j|^{-1}$ (with a suitable alteration at $i = j$).

Simple grids like this are rarely used for electronic structures because N for an accurate 3D grid would be much bigger than for a typical basis. Consider a single atom: the grid spacing needs to be set to resolve the behavior near the nucleus, resulting in many grid points to describe the tails of the wavefunction far from the nucleus. In contrast, in a Gaussian basis, the tails of the wavefunction can be described using just a few basis functions. For example, in an early effort by the author using a uniform 3D grid, utilizing finite elements rather than finite differences, as many as 10^5 grid points were needed for accurate results for the molecule.[6] Much more efficient approaches use adapted grids, putting more points near the nuclei.[4] However, it is difficult to make the number of grid points nearly as small as the number of basis functions in a Gaussian-type basis. Wavelet bases are another approach with much of the locality of a grid and with adaptable increased resolution near nuclei, but again the number of functions tends to be substantially larger than that in Gaussian bases. The larger number of functions is a particular problem for methods which treat correlation especially accurately.

Grid-like methods have a fundamental advantage for use with DMRG and other recently developed tensor network methods: these methods are based on the low entanglement of ground states when expressed in a local real-space basis. The entanglement of ground states is governed by the *area law*,[14,15] which is specific to localized real space bases. In a delocalized basis, a volume law of entanglement holds instead, requiring the number of states m kept to grow exponentially for fixed accuracy. Thus the DMRG greatly prefers a real-space local representation. In addition, the calculation time for the DMRG depends strongly on the number of two-electron interaction terms, so the N^4 scaling of V_{ijkl} is a major drawback. The standard DMRG approach for molecules in a Gaussian basis utilizes a standard basis localization method before the DMRG is used. This localization is less than ideal, and except in certain treatments of long chains, all N^4 two-electron terms are kept.

The very recently developed sliced basis DMRG (SBDMRG)[16] approach uses a finite-difference grid in one direction (the long direction in a chain) and 2D Gaussians for the transverse directions. For chain systems, the SBDMRG grid gives a reduction to $O(N^2)$ two-electron terms, where N scales linear with the length. SBDMRG also utilizes a compression algorithm for the long range terms, yielding an $O(N)$ calculation time, and chains of up to 1000 hydrogen atoms have been studied in the strongly correlated stretched-bond regime. While arbitrary molecules can, in principle, be treated with SBDMRG, it is particularly suited for long chains, even more than the standard quantum chemistry DMRG. One motivation for this work is to find a good way to increase the grid spacing in the SBDMRG without loss of accuracy, for faster and more efficient SBDMRG calculations. However, a more important motivation is to take advantages of locality in all three dimensions.

Here we present techniques that do take us closer to an ideal combination of grid and basis discretizations. Our approach is related to orthogonal wavelet bases. Wavelets have and continue to be used successfully in electronic structure calculations, but our goals are different from many existing uses. A conventional wavelet basis can be used in a way that has precise control over the accuracy, with systematically improvable accuracy, in, say, a density functional theory calculation.[1] These approaches make no use of standard Gaussian bases which give an excellent description of core electrons with a very small number of functions. Thus the price of the precise control of accuracy is a basis that may have more than 1000 functions for a small molecule, which is not practical for most beyond-HF correlation calculations.

Our interest is in these correlation calculations, where one is interested in accuracies only up to about 1 mH and for which Gaussian basis set descriptions of cores are fine, at least for energy differences. Thus we consider approaches where we use wavelet techniques in a very restricted way, combining them with standard Gaussian basis sets. A key consideration in developing such an approach is the calculation of integrals. To make this easy, we construct functions similar to the scaling functions of wavelet transforms, but out of an equal-spaced array of identical-width Gaussians. The resulting functions called *gausslets* have a number of significant advantages for electronic structure calculations.

Gausslets live on a uniform grid, with one function per grid point. They are defined in 1D, but in 2 or 3D, one simply takes products of the 1D functions. Gausslets are orthogonal and symmetric and almost compact in both real and momentum space, in the same sense that Gaussians are. They have excellent completeness properties for representing smooth functions. All standard integrals have simple analytic forms although one has to contract over the underlying Gaussians. A gausslet is shown in Fig. 1, together with seven of the Gaussians of which it is composed. A 1D gausslet basis would be composed of this function plus integer translations, and the whole basis can be scaled to a grid spacing a.

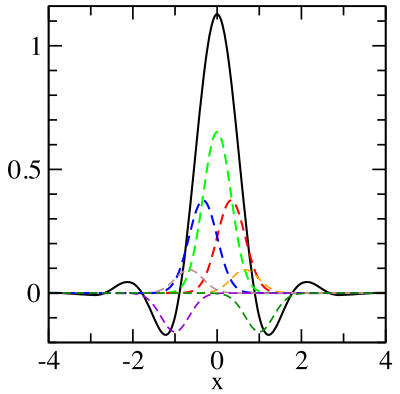

FIG. 1. The gausslet $\mathcal{G}_6$ (solid line), along with some of the Gaussians of which it is composed, multiplied by their coefficients. A basis is formed by translating $\mathcal{G}_6$ to center it on every integer and then scaling to any desired lattice spacing a (here $a = 1$). This basis is orthonormal and can represent any polynomial up to fourth order. Gausslets are exactly symmetric and are defined as a sum of Gaussians of width 1/3, with spacing 1/3; $\mathcal{G}_6$ falls to about 10^{-12} at $x = \pm 12.5$.

Gausslets are constructed using a recently developed class of wavelet transforms which use a factor of three-scaling transformation rather than the usual factor of two, in order to make them exactly symmetric. They have an additional very important property: they integrate like a δ-function, for any low-order polynomial $p(x)$,

$$\int_{-\infty}^{\infty} dx\, \mathcal{G}(x-b)p(x) = p(b). \tag{2}$$

This, in turn translates to reducing the two-electron integrals from N^4 to N^2, $V_{ijkl} \to \tilde{V}_{ij}$, provided the interaction is smooth. One way to make the interaction smooth would be to replace the Coulomb electron–electron interaction with a two-electron pseudopotential.[17]

A motivation for the gausslet development was to make a basis that was easy to use in combination with standard Gaussian bases. In particular, if one thinks of a standard grid as being defined by finite differences, it is not at all clear how one can add, say, an extra atom-centered narrow-width Gaussian to the representation to describe a nuclear cusp. But as soon as one chooses to attach basis functions to each grid point, then it becomes clear how one can add additional non-grid functions. This would be true also for a grid of sinc functions, and with FFT methods one can do integrals with sinc functions quickly. However, we also wish to orthogonalize the combined basis (where either gausslets or sinc functions are orthogonal among themselves). Ideally, we would also want to be able to orthogonalize separately around each atom. This gives very localized bases, such as gausslets, an advantage. The number of gausslet-Gaussian integrals is reduced by the locality, and their evaluation is made easier by the fact that the gausslets are composed of Gaussians. These practical issues involving combining Gaussians and gausslets, and doing the needed integrals, are important but are beyond the scope of the current paper.

The first step in constructing gausslets is understanding the properties of arrays of equally spaced, equal-width Gaussians.

III. ARRAYS OF GAUSSIANS

Arrays of Gaussians with identical widths are particularly convenient to use in constructing basis functions because of their analytic integrals, their smoothness, their completeness, and also because the product space is greatly reduced and convenient. The product of a Gaussian of width w located at i, and another of width w at j, is a Gaussian centered at $(i+j)/2$ of width $w/\sqrt{2}$; the set of all products of a grid of N such Gaussians is a half-spaced grid with roughly $2N$ functions, rather than N^2 functions. The product space plays a central role in defining Hamiltonian matrix elements, so this simplification can significantly improve computational efficiency. (Polynomials also have similar features of their product space, namely, that a product of two-order N polynomials is an order $2N$ polynomial. This representation is used in the exact tensor hypercontraction approach.[13])

The completeness properties of arrays of Gaussians are interesting. As is common in numerical analysis, we define completeness in terms of representing low order polynomials. A grid of basis functions with good polynomial completeness is excellent at representing arbitrary smooth functions. Consider a unit-spaced 1D grid, and on each integer point j, put a Gaussian with width w,

$$g_j(x) \equiv \exp[-\frac{1}{2}\frac{(x-j)^2}{w^2}]. \tag{3}$$

To see the completeness of the set $\{g_j\}$, consider the sum, shown in Fig. 2(a),

$$C(x) = \sum_j g_j(x). \tag{4}$$

Since g_j are identical except for translation, the representation within this basis of a constant function must be proportional to $C(x)$. If $C(x)$ is not nearly constant, the basis is very poor—it cannot even represent a constant.

We will show that $C(x)$ is very nearly constant for sufficiently large w. Note that C is periodic, $C(x-1) = C(x)$, with period 1. Expand it in a Fourier series, with coefficients $c(k = 2\pi n)$, where n is an integer

$$\begin{aligned} c(k) &= \int_0^1 dx\, C(x) \exp(-ikx) \\ &= \sqrt{2\pi}\, w \exp(-\frac{1}{2}k^2 w^2). \end{aligned} \tag{5}$$

The largest deviation from constancy comes from the smallest nonzero k in the series, namely, $k = \pm 2\pi$, for which the factor

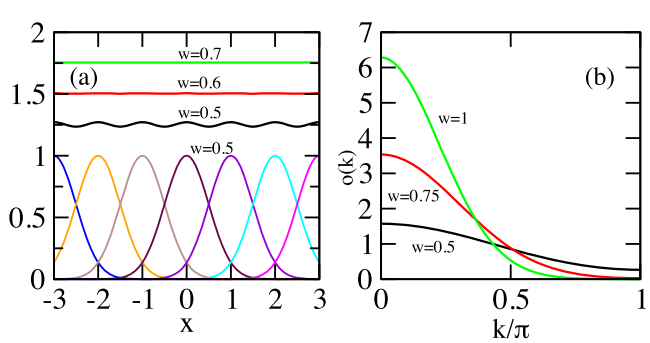

FIG. 2. (a) An array of Gaussians and their sums, $C(x)$. (b) Overlap matrix eigenvalues.

$\exp(-\frac{1}{2}k^2w^2)$ is $\exp(-2\pi^2w^2)$. For $w = \sqrt{2}$, this factor is less than 10^{-17}. For $w = 1$, it is less than 10^{-8}. Thus for modest width w, $C(x)$ is exactly constant to double precision accuracy (15 or 16 digits). In what follows, we choose $w = 1$ since this lack of completeness at the 10^{-8} level appears in the wavefunction. According to standard variational arguments, it translates to errors in the energy of order 10^{-16}. Note that higher terms in the series have much smaller coefficients so that not only $C(x)$ is nearly constant but also a number of higher derivatives are nearly zero. We will not try to make this particularly quantitative since we can test any such property with a trivial numerical evaluation.

The flatness of $C(x)$—for the special case of Gaussians—also implies higher order polynomial completeness. In particular, we will show that the near constancy of $C(x)$ implies that, for $n \geq 1$, $\sum_j j^n g_j(x)$ is very nearly a polynomial in x with order n. (We have already shown it is true for $n = 0$.) If *some* linear combination of g_j can represent an order-m polynomial for $0 \leq m \leq n$, then one can find a suitable linear combination that can represent *any* polynomial up to order n. Gaussians have the property that their derivatives, to all orders, can be written as the same Gaussian times a polynomial. In particular, let $P(x, n)$ denote a polynomial in x of degree n; the nth derivative of $g_j(x)$ can be written as

$$\begin{aligned} g_j^{(n)}(x) &= P(x-j,n)g_j(x) \\ &= \sum_{l,m} C_{l,m} j^l x^m g_j(x), \end{aligned}$$

where $C_{l,m} = 0$ if $l + m > n$. The nth derivative of $C(x)$ is

$$0 \approx C^{(n)}(x) = \sum_{l,m} C_{l,m} x^m \sum_j j^l g_j(x). \tag{6}$$

In order to prove our result by induction, we pick out the $l = n$ term, for which $m = 0$, and assume that for $l < n$, $\sum_j j^l g_j(x)$ is a polynomial $P_l(x)$. Then

$$C_{l,0} \sum_j j^n g_j(x) \approx -\sum_{l<n,m} C_{l,m} x^m P_l(x). \tag{7}$$

Since $C_{l,m} = 0$ if $l + m > n$, this means that if $\sum_j j^l g_j(x)$ is very nearly an order-l polynomial for all $l < n$, then it is an order n polynomial for $l = n$. By induction, this is true for any n. This result breaks down when $C^{(n)}(x)$ stops being zero for large n. For $w \gtrsim 1$, the breakdown does not occur very quickly, and an array of Gaussians has approximate polynomial completeness to high order.

The weakness of a grid of Gaussian as a basis is its lack of orthogonality, and that orthogonalizing the functions involves a fairly singular matrix, associated with a near lack of linear independence. Let $S(j, k) = S(j - k) = \langle g_j | g_k \rangle$ be the overlap matrix of $\{g_j\}$. We can form an orthonormal set of functions (all related by integer translations) using $q(j - k) \equiv S^{-1/2}(j, k)$ as

$$G_j(x) \equiv \sum_k q(j-k)\, g_k(x). \tag{8}$$

However, S becomes increasingly singular as w increases. The eigenvectors of S are plane waves, and the most singular point is at momentum π; this corresponds to near cancellation of g_j. For example, for $w = \sqrt{2}$, the unnormalized fit to a momentum π plane wave, $\sum_j (-1)^j g_j(x)$, is only of order 10^{-4} in magnitude. The near singularity of S means that G_j have long tails and widely varying coefficients for the component Gaussians, making them computationally poorly behaved.

Fortunately the orthogonalization does not spoil the completeness nor would a generic convolution of the form Eq. (8), transforming $g \to G$ using an arbitrary vector q. Consider

$$\sum_j P(j,n)G_j(x) = \sum_k \left[\sum_j q(j-k)P(j,n) \right] g_k(x). \tag{9}$$

The term in square brackets is a polynomial in k of degree n, so the entire right side is a polynomial in x of degree n. By the same reasoning as above, this implies that G_j have the same excellent completeness properties as g_j.

In summary, a grid of Gaussians has excellent completeness, as well as several other very desirable features, but poor orthogonality and linear independence properties. Fortunately, as we show in Secs. IV and V, these faults can be fixed using wavelet technology.

IV. TERNARY WAVELET TRANSFORMATIONS

In conventional compact orthogonal wavelet theory, as pioneered largely by Daubechies,[18] it is not possible to have a symmetric, compact, orthogonal wavelet transformation (WT). However, with ternary WTs, which change scales by a factor of 3 instead of 2, symmetry, orthogonality, and compactness are compatible. Recently Evenbly and White (E and W)[19] introduced new families of ternary WTs with excellent completeness, compactness, and smoothness properties, which we will make use of and extend. A very carefully chosen set of coefficients c_k, $-m \leq k \leq m$, with $c_{-k} = c_k$, defines a symmetric wavelet transform. Given a single function $f(x)$, we define all integer translations $f_j(x) = f(x - j)$ to form a basis set that lives on an integer grid. Then the wavelet transform produces a new basis set, also living on a grid with unit spacing, defined by a function f', defined by

$$f'(x) = \sum_k c_k f(3x - k). \tag{10}$$

The scaling function of the WT is the fixed point of Eq. (10). The fixed point exists provided the initial f satisfies some simple conditions, most importantly that they sum to a constant (exactly), $\sum_j f_j(x) = \text{const}$. Thus any functions defined by Gaussians with finite w are technically not suitable for generating the fixed point. However, if one only wants to apply the WT a modest number of times, Gaussian derived functions can be an excellent starting point. Since we do not plan to use the wavelets to represent sharp core functions (instead using standard Gaussians), a full multiscale resolution analysis is not needed, and for our uses, the fixed point is entirely unnecessary.

E and W showed that WTs have a direct correspondence with quantum circuit theory. The circuit corresponding to a WT is defined by a small number of angles θ_k; each angle defines a unitary "gate," which is a 2×2 or 3×3 unitary matrix. The unitarity of the circuit is independent of the values of θ_k, removing an annoying set of constraints when optimizing a WT. The circuits express symmetry much more naturally than in conventional approaches. Some of the symmetric ternary

WTs constructed by E and W appear to have better properties than any previous such WTs.

Besides the transformation of the scaling function in Eq. (10), the WT produces two wavelet functions per scaling function. The scaling function captures the lowest momentum behaviour, while the wavelets describe high momentum. These wavelets are less central to our discussion here.

In the terminology of E and W, we consider here only type I site-centered symmetric ternary wavelets. These are characterized by the number of low and high frequency moments associated with the scaling functions. We make this more specific here with the notation W_{nlh} to describe the WT with n angles, l low moments, and h high moments. The number n also gives the number of layers of the circuit, fixing the range of c_k, specifically from $-(3n-2)$ to $3n-2$, with $c_{-k} = c_k$. The number l gives the completeness of the scaling functions; more specifically, $l-1$ is the maximum polynomial order the scaling functions fit exactly, which is imposed by making the wavelets orthogonal to any lth degree polynomial. Similarly, h controls the smoothness; specifically, $h-1$ gives the number of sign-flipped polynomials $(-1)^k k^h$ that the scaling functions are orthogonal to; it also means that, the Fourier transform of c_k has vanishing value and $h-1$ vanishing derivatives at maximum momentum π.

A related and important property of the corresponding scaling functions is that they integrate polynomials like a δ-function. This is associated with a property of c_k,

$$\sum_k c_k k^m = \sqrt{3}\delta_{m,0} \tag{11}$$

for $m = 0, \ldots, p-1$ for some p. (The evenness of c_k makes this statement trivial for odd m.) The δ-function order p is related to the other moments but not in a simple way: we have found WTs with the same n, l, and h, but with different p. Usually, however, a large p appears "for free" from optimizing l and h.

E and W give just a few examples of these types of WTs and not to full double precision. Determination of W_{nlh} involves a nontrivial nonlinear optimization; we have found W_{nlh} to high precision both for the examples of E and W and a number of additional WTs with higher order. All useful WTs had even n, and it is difficult to converge useful WT for higher n. Angles for the most useful WTs are listed in Table I. These emphasize completeness over smoothness, but not completely, mostly setting $h = 2$ (which is only one constraint since the order-1 high moment is automatically satisfied because of symmetry). The scaling functions with $h = 2$ are nicely smooth; for $h = 0$, they are much more irregular. This leaves $n-1$ degrees of freedom for completeness, i.e., $l = n-1$. Below we construct gausslets out of W_{432}, W_{652}, W_{872}, and W_{1092}. These WTs all have impressively high p's given by a simple formula: $p = 2l$.

A WT can be applied to an array of Gaussians, producing a new basis where the functions are formed from sums of Gaussians. The completeness properties of the Gaussians are transferred by the WT to the new basis, up to the completeness order l. To see this, let i index a grid with spacing 1, and j (an integer) run over a grid with spacing 1/3. Let $f_i(j) = c_{j-3i}$; we can think of f_i as being a discrete basis function representing functions living on the 1/3 grid. (Usually in wavelet theory, one thinks of an infinite sequence of ever smaller grids, which represents the continuum in the limit of infinitesimal grid spacing. In contrast, we apply the WT to a grid of functions which are already continuous functions of x.) The WTs have a discrete completeness property that means we can write

$$j^m = \sum_i a_i f_i(j) \tag{12}$$

for $m < l$. Multiplying by $g_j(x)$ on both sides and summing over j, we find that linear combinations

$$s_i(x) \equiv \sum_j f_i(j) g_j(x) \tag{13}$$

can represent arbitrary low order polynomials. Again, this implies that suitable linear combinations of $s_i(x)$ can represent any low order polynomial. However, they are not orthogonal, which we address in Sec. V.

TABLE I. Angles θ_k parameterizing wavelet transforms of various depths.

	W_{212}	W_{220}
θ_1	0.169 918 454 727 060 968 55	0.275 642 799 216 265 403 97
θ_2	0.785 398 163 397 448 309 62	0.679 673 818 908 243 874 19
	W_{432}	W_{652}
θ_1	0.335 914 092 490 436 356 38	0.478 080 355 350 786 627 12
θ_2	−1.469 776 795 453 469 694 82	0.617 246 034 083 187 914 23
θ_3	−0.165 995 637 763 375 387 84	−1.395 269 399 144 701 110 11
θ_4	2.255 174 958 850 918 004 43	−0.514 530 804 781 996 704 77
θ_5	...	1.087 107 498 520 975 451 54
θ_6	...	0.682 682 934 096 257 100 15
	W_{872}	W_{1092}
θ_1	0.575 486 325 541 892 993 96	0.611 838 641 007 119 145 17
θ_2	1.070 928 966 976 834 574 30	0.706 801 895 273 396 593 67
θ_3	−0.533 970 487 578 274 787 00	−2.760 902 224 387 028 914 61
θ_4	−0.844 040 574 903 297 846 56	0.827 251 757 908 607 674 89
θ_5	−2.197 797 947 694 209 725 67	−0.634 937 861 307 327 875 50
θ_6	−0.209 022 939 514 514 817 99	−0.233 462 171 969 760 425 71
θ_7	2.326 200 564 457 652 486 95	1.213 278 913 489 122 940 41
θ_8	0.767 532 710 838 426 399 87	−1.153 901 815 978 609 033 52
θ_9	...	1.740 640 985 925 175 673 08
θ_{10}	...	0.638 708 498 163 813 500 29

V. GAUSSLETS

We wish to use these properties of WTs and Gaussian grids to define convenient, orthogonal, complete functions defined as sums of Gaussians. There are several possible ways to proceed. One way starts with $s_i(x)$. $s_i(x)$ are not orthogonal, but their overlap matrix S_s is much less singular than that of $g_j(x)$. $s_i(x)$ look like somewhat broadened wavelet scaling functions, with oscillations that partially cancel off-diagonal elements of the overlap matrix. (Further applications of a WT would take them closer to orthogonality.) One can use $S_s^{-1/2}$ to orthogonalize s_i. The resulting functions are close to what we want, but $S_s^{-1/2}$ introduces moderately long tails in the final functions. One can then apply a WT to these orthogonal functions, which would substantially decrease the tails, and use those functions. The main drawback is that the underlying Gaussian grid would

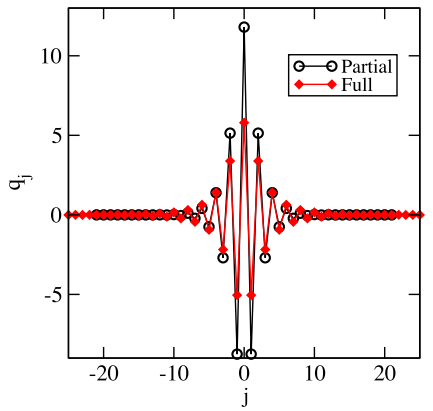

FIG. 3. Full and partial orthogonalizing vectors $q(k)$, for the case of $\mathcal{G}_6$. The partial orthogonalizer extends out to ±21; the full orthogonalizer has fallen to about 10^{-12} at $j = 110$.

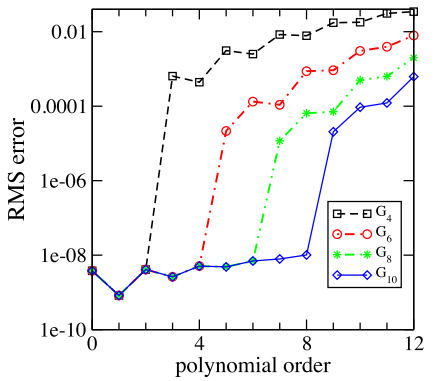

FIG. 4. Root mean square errors in the fit of an array of gausslets with unit spacing to scaled Chebyshev polynomials $T_n(x/10)$, over the range $0 \le x \le 1$, for gausslets of different orders, as a function of n. The gausslets of order n are based on wavelet transformations designed to fit polynomials of order 0 to $n-2$. The residual error of order 10^{-8} visible in the figure for small n is due to the finite width and imperfect completeness of the underlying Gaussians making up the gausslets. Since the underlying Gaussians are the same for all the gausslets, the residual errors are approximately the same. The fitting errors of order 10^{-8} are expected to translate to energy errors of order 10^{-16}.

have a spacing of 1/9 that of the final basis functions. The more the Gaussians making up a basis function, the more is the computational work to use them, so we have developed a way to maintain the minimal 1/3 spacing of the underlying Gaussians without the long tails.

It is possible to partially orthogonalize $g_j(x)$ so that the application of a particular WT to the partially orthogonalized functions produces fully orthonormal functions. Let function $G_j(x)$ be defined by Eq. (8), for some vector $q(k)$, and then apply a WT [defined by $f_i(j)$],

$$\mathcal{G}_i(x) \equiv \sum_j f_i(j)\, G_j(x). \tag{14}$$

We know that $q(k)$ exists that makes the $\mathcal{G}_i(x)$ orthonormal—we can use the overlap matrix S_s to make $q(k)$, which completely orthonormalizes $G_j(x)$, and thus $\mathcal{G}_i(x)$. In that case, $q(k)$ falls to about 10^{-12} near $k = 110$, so it is rather nonlocal, making $\mathcal{G}_i$ have long tails. It is possible to find a much more compact $q(k)$ if it is optimized for a particular WT, requiring orthogonality only of $\mathcal{G}_i$ and not both G_j and $\mathcal{G}_i$. The q convolution and the WT, each take care of parts of the orthogonalization. A typical q is shown in Fig. 3. Finding such q is a rather tedious nonlinear optimization problem, with many local minima. We implemented this optimization using a Nelder Mead algorithm, using restarts and high precision to deal with local minima, with the minimizations sometimes taking several days, and only achieving approximate double-precision orthogonality. A final orthogonalization step was performed to make the functions orthogonal to very high precision, with minimal impact on locality, using the inverse square root of the overlap matrix. Fortunately, using these functions does not require that these optimizations be repeated; all one needs are the final coefficients of the Gaussians, and the properties of the functions are easily verified.

This approach produces functions—gausslets—with a ratio of 3 between the spacing of the functions and the underlying Gaussians. Normally with WTs, there is a tradeoff between order and compactness. The presence of the underlying Gaussians means that there is an inherent limit to the size of a "1/3-gausslet" which is greater than the width of the fixed point scaling function of the WT used to make the gausslet. However, the gausslets are substantially smoother. We present coefficients for gausslets $\mathcal{G}_4$, $\mathcal{G}_6$, $\mathcal{G}_8$, and $\mathcal{G}_{10}$, where the index signifies n of the underlying WT, W_n, which has $l = n - 1$ and $h = 2$. The gausslet $\mathcal{G}_6$ is shown in Fig. 1; the others look similar, with slightly increasing width with the order. The coefficients b_j of these gausslets in terms of the underlying Gaussians are given in the Appendix in Tables II–V.

A simple test of the completeness of the gausslets is shown in Fig. 4. The gausslets are fitted to polynomials of various orders and the root mean square errors over the unit interval are shown. (The gausslets involved in the fitting extend well outside this interval because of the small but finite tails of the gausslets.) The fitting coefficients were approximated using the δ-function property, as $T_n(i/10)$ for the gausslet centered at i, so this test of the fitting also tests the δ-function property. Since the order of the δ-function property is much higher than for fitting property, one sees only the errors associated with the fitting.

One may wish to have more compact gausslets, paying the price of an underlying finer array of Gaussians. Suitable 1/9, 1/27, etc., gausslets are easily derived by applying WTs to one of the $\mathcal{G}_n$. We label these gausslets as in the following example: $\mathcal{G}_{864}$ is formed from $\mathcal{G}_8$ and applying W_6 and then W_4. These constructions also generate useful wavelet-like functions, and a multiresolution basis can be formed for a modest range of scales, with each length scale having different functions. If one successively applies W_n, the range of the functions decreases, and the basis functions become close to the scaling function of W_n. Alternatively, one can successively decrease the order over several steps, which produces very localized functions which are still smooth with a finite size underlying Gaussian array. An example of this is shown in Fig. 5. A multiscale basis can be useful, but in many cases, a better alternative that produces smaller bases is to add atom-centered Gaussian functions to the gausslet basis. The extra functions can be orthogonalized to the gausslets and themselves. A simple 1D example of this approach is given below.

As a first test of the use of gausslets to solve the Schrödinger equation, we apply them to two different simple

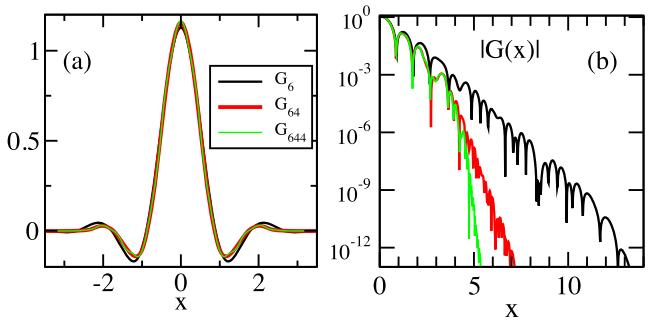

FIG. 5. Comparison of a sequence of gausslets $\mathcal{G}_6$, $\mathcal{G}_{64}$, and $\mathcal{G}_{644}$. Repeated application of a wavelet transform shrinks the tails of the functions.

1D smooth potentials; see Fig. 6. The first, the Pöschl-Teller potential, is exactly solvable, with energy $E = -1/2$. The second, the soft Coulomb (SC) potential, has been used as a 1D model with properties such as correlation energies roughly similar to that of real 3D molecules.[20,21] Here the SC potential would be that of pseudo-hydrogen. Each potential is centered at two positions to demonstrate that the grid points need not be aligned with the potential centers. To define the Hamiltonian matrices, we first evaluate the Hamiltonian matrix elements for the underlying Gaussian grid with spacing $a/3$, analytically for the kinetic energy and numerically for the potentials. Then we transform to the gausslet basis using b_j. Both of these potentials have widths of order 1, and we see that very high accuracy is easily achievable for $a \sim 0.2$. More important for our goals is that one achieves accuracies of $O(10^{-3})$ for $a \sim 1$. Large a values are essential for applications to 3D.

A key advantage of a basis set approach is that one can add functions to it to represent singularities or other sharp features. As a test of this, shown in Fig. 7, we consider a potential with a smooth part plus an off-center delta function, $V(x) = -\exp(-\frac{1}{2}x^2) - \delta(x - 2.7)$. Particularly in three dimensions, this sort of potential is difficult with a grid. Our basis consists of an array of gausslets plus, to represent the singularity, the solution to the delta function alone, $\exp(-|x - 2.7|)$. The exponential is orthogonalized to the gausslets to make a fully orthogonal basis. To carry out the calculations, the

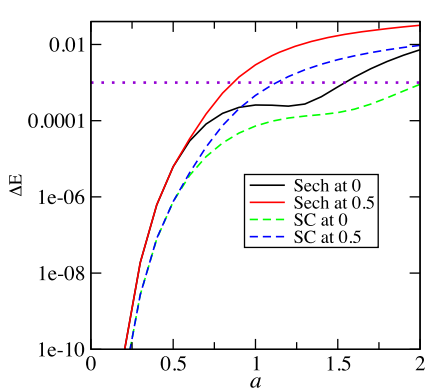

FIG. 6. Energy errors in solving 1D potentials with a $\mathcal{G}_{10}$ gausslet basis with spacing a. Here the "Sech" potential is the exactly solvable Pöschl-Teller potential $V(x) = -\text{sech}^2(x - b)$, for $b = 0, 0.5$, and the "SC" potential is the soft Coulomb potential $V(x) = -1/\sqrt{(x-b)^2 + 1}$, for $b = 0, 0.5$. The dotted line shows $\Delta E = 10^{-3}$, a typical target accuracy for "chemically accurate" electronic structure calculations.

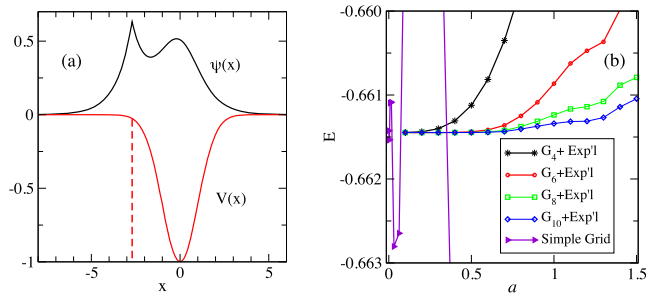

FIG. 7. (a) A potential with a smooth part and a delta function, and the resulting ground state. (b) Convergence of the energy with grid spacing for two types of bases: (1) an array of gausslets plus an exponential which is the solution of the isolated delta function, for four different gausslets, and (2) a simple second order finite difference grid.

exponential is first represented very accurately as a sum of a few hundred Guassians, the coefficients coming from a discretization of an integral. (In a 3D calculation, to describe, say, a 1S function, one would use a sum of a few Gaussians from a standard basis set.) Then all the potential terms are sums of analytic Gaussian integrals. The hybrid Gaussian/exponential approach converges very rapidly, with errors below about 10^{-3} for $a < 1.5$ with $\mathcal{G}_8$ and $\mathcal{G}_{10}$. The energies for the last two points, $a = 0.1$ and $a = 0.2$, agree to eight digits, $E = -0.66\,144\,716$. For comparison, we implemented a naive grid, without trying to put a grid point exactly on the singularity. The grid results show substantial oscillations as a function of a. The finest grid result, $a = 10^{-4}$, gives $E \approx -0.6614$, in agreement with the gausslet result but much less accurate.

VI. DIAGONAL APPROXIMATIONS

A crucial feature of finite difference grid approximations is the simple form of the two-electron interaction, $V_{ij}\hat{n}_i\hat{n}_j$. A similar property holds for the one electron nucleus-electron potential terms, which have the simple form $U_i\hat{n}_i$. This latter property is not so important in itself since the extra computation time to deal with a U_{ij} form of the potential is minor, but it does serve another useful purpose: approximations which make U diagonal should generally do the same for V since the two-electron interaction acts on one electron with a single-particle potential determined by the locations of all the other electrons. We will consider the simpler single particle potential first, in one dimension. Diagonal representations such as these are a key property of bases made from the sinc function.[7] Here we find similar approximations for gausslets, implying that the high nonlocality of the sinc is not needed for this sort of approximation. Although we focus on the gausslets, similar approximations would work for the ternary wavelet scaling functions and for certain types of traditional wavelet scaling functions with the δ-function property, particularly coiflets.[18] Devising methods to reduce V_{ijkl} to matrices or tensors of size $O(N^2)$, using tools such as density fitting and tensor factorization, is also a central focus of other modern approaches.[12,13]

Let $U(x)$ act on a continuum wavefunction $\psi(x)$ to give another wavefunction $\phi(x)$: $\phi(x) = U(x)\psi(x)$. We are interested in the coefficient of ϕ for basis function $\mathcal{G}_i$, written using the expansion of ψ in terms of $\mathcal{G}_j$,

$$\phi_i = \int dx\, \mathcal{G}_i(x)\phi(x) = \sum_j \psi_j \int dx\, \mathcal{G}_i(x)U(x)\mathcal{G}_j(x). \quad (15)$$

This is the conventional non-diagonal form, with the last integral defining U_{ij}. Now assume that each $\mathcal{G}_i$ integrates like a weighted delta function, with location x_i,

$$\int dx\, \mathcal{G}_i(x)f(x) = f(x_i)w_i \quad (16)$$

for any smooth function $f(x)$, with

$$w_i \equiv \int dx\, \mathcal{G}_i(x). \quad (17)$$

Then

$$\phi_i = \phi(x_i)w_i = U(x_i)\psi(x_i)w_i = U(x_i)\psi_i. \quad (18)$$

Thus we have a diagonal "point" approximation for U,

$$U_{ij} \to \delta_{ij}U(x_i). \quad (19)$$

Note that the locality of the gausslets means that U_{ij} is small for i far from j, but the diagonal approximation does not rely on this. In particular, we disregard the near-neighbor $j = i \pm 1$ terms even though they are not small.

There are two other closely related diagonal approximations. First, if the δ-function property is only approximate, we may prefer to replace $U(x_i)$ with its overlap with $\mathcal{G}_i$,

$$U_{ij} \to \delta_{ij} \int dx\, \mathcal{G}_i(x)U(x)/w_i. \quad (20)$$

One might hope that this "integral" approximation could be more accurate since it averages the potential over a finite range, but one needs tests to see if it actually is an improvement. Second, let us assume that $\{\mathcal{G}_i\}$ can exactly represent a constant function over the range of interest. Then w_k is then the expansion coefficient for $\mathcal{G}_k$ to represent the identity function, and

$$1 = \sum_k w_k\mathcal{G}_k(x). \quad (21)$$

Inserting this into Eq. (20) gives the "summed" approximation

$$U_{ij} \to \delta_{ij} \sum_k U_{ik}w_k/w_i. \quad (22)$$

For the special case where $\mathcal{G}_i$ are a uniform grid of gausslets, $w_k = w_i$, and as the grid spacing goes to zero, smooth functions have $\psi_k \approx \psi_i$ in the close vicinity of a point i. In this case, $\sum_k U_{ik}\psi_k \approx \psi_i \sum_k U_{ik}$, providing another rough justification for Eq. (22). (This idea was the original inspiration for these diagonal approximations.)

Tests of the diagonal approximations are shown for the soft Coulomb potential in Fig. 8. One sees that all of the diagonal approximations become very precise for small a. In fact, although one cannot see it in this figure, the diagonal approximation curves and the non-diagonal full matrix curve come together faster as a function of a than any of the curves converge to the exact result. This is because the δ-function property parameter p is twice that of the completeness parameter l. This means that at small a, there is no point in using the full matrix—the result might not be exact, but the diagonal approximation does not contribute significantly to any error. At larger a, the diagonal approximations introduce significant errors. The integral approximation (or equivalently for this case, the summed approximation) is significantly better than the point approximation.

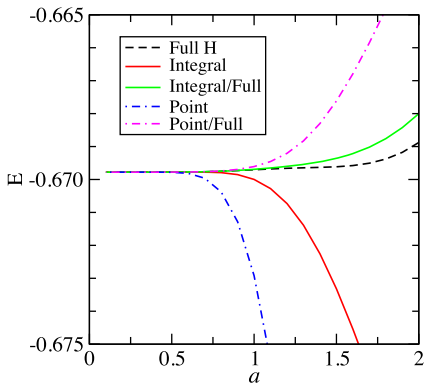

FIG. 8. Energy solving the soft Coulomb potential (with $b = 0$) using diagonal approximations with the $\mathcal{G}_{10}$ gausslet basis, as a function of spacing a. The "full" curve uses the full Hamiltonian matrix. The "integral" diagonal approximation is shown, which for these bases is identical to the "summed" approximation. For the two diagonal approximations, we also show the energy evaluated with the full Hamiltonian, but with the eigenvector from the corresponding approximation.

These three approximations translate immediately into two-electron approximations for V. The pointwise evaluation is

$$V_{ijkl} \to \delta_{il}\delta_{jk}V(x_i, x_j). \quad (23)$$

The integral approximation replaces $V(x_i, x_j)$ in Eq. (23) with

$$V(x_i, x_j) \to \int dx\, dx'\, \mathcal{G}_i(x)V(x, x')\mathcal{G}_j(x')/(w_i w_j). \quad (24)$$

The summed approximation replaces $V(x_i, x_j)$ with

$$V(x_i, x_j) \to \sum_{kl} V_{ijkl}w_l w_k/(w_i w_j). \quad (25)$$

Because of the big difference in computational scaling associated with two-electron versus one-electron terms, it is sensible to only make the diagonal approximation for the two-electron terms, using the full matrices for the one electron terms.

As a simple interacting example calculation, we perform an exact diagonalization for a 1D helium atom in a soft Coulomb potential, with a basis of gausslets ranging from $-L$ to L. The results are shown in Fig. 9. According to Ref. 21, the

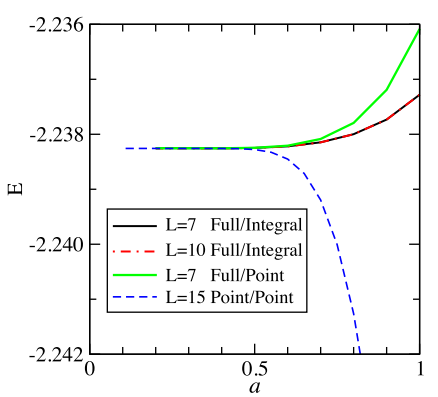

FIG. 9. Energy solving the soft Coulomb 1D helium atom using gausslet basis $\mathcal{G}_{10}$ with diagonal approximations, as a function of spacing a. The approximations are labeled in the form (single-particle approximation)/(two-particle approximation). The lattice of gausslets extends approximately from $-L$ to L.

exact energy from a grid DMRG calculation is −2.238. The two full/integral curves, which completely overlap, demonstrate that $L = 7$ gives excellent convergence in L. All the diagonal approximations perform excellently at smaller a. For larger a, they still behave well; the full/integral approximation stays within 1 mH almost up to $a = 1$. The point diagonal approximations are somewhat less accurate at larger a, but the point/point approximation is especially convenient for small a since it is as accurate as the other approximations and requires no integral evaluations at all. For this case, we consider $a = 0.1$ and $L = 15$, which gave 601 basis functions, to push for very high accuracy; we obtain $E = -2.238\,257\,824$, which we believe is correct for all the digits shown. The setup and diagonalization took less than 5 min on a desktop. With DMRG, it would be straightforward to extend these calculations to long chains.

One can consider making the diagonal approximations after first making a mean-field-like reduction of the operator. For the one electron case derived below, this gives an energy correction but does not change the Hamiltonian. For the two-electron terms, this produces altered one electron terms which change the diagonalization and might improve the results. For the one electron case, assume we have some estimate for $\langle \hat{c}_i^\dagger \hat{c}_j \rangle$, and let

$$\hat{U} = \sum_{ij} U_{ij} \hat{c}_i^\dagger \hat{c}_j \tag{26}$$

$$= \sum_{ij} U_{ij} (\hat{c}_i^\dagger \hat{c}_j - \langle \hat{c}_i^\dagger \hat{c}_j \rangle) + \sum_{ij} U_{ij} \langle \hat{c}_i^\dagger \hat{c}_j \rangle. \tag{27}$$

If we now apply a diagonal approximation $U_{ij} \to \tilde{U}_{ii}$ on the first term, we have

$$\hat{U} \approx \sum_i \tilde{U}_{ii} \hat{c}_i^\dagger \hat{c}_i + \left(\sum_{ij} U_{ij} \langle \hat{c}_i^\dagger \hat{c}_j \rangle - \sum_i \tilde{U}_{ii} \langle \hat{n}_i \rangle \right). \tag{28}$$

Suppose we find the ground state using $\tilde{U}$ and from this compute $\langle \hat{c}_i^\dagger \hat{c}_j \rangle$ and $\langle \hat{n}_i \rangle$, and then plug this into Eq. (28). Then adding in the correction term in the parentheses in Eq. (28) is the same as evaluating the energy in the full Hamiltonian of the eigenstate from the approximate Hamiltonian. This corrected result is shown in Fig. 8. We see that the corrected result is significantly better than the uncorrected result at large a. We leave exploring these mean-field approximations for the two-electron term for future work.

A basis permitting a good diagonal approximation is special, and an orthogonal transformation of the basis will, in general, spoil the approximation. A truncation of the basis may not harm the approximation; in particular, a reduction of a particular set of orbitals into *one* orbital which is a linear combination from the set still allows a diagonal approximation. This is because a basis rotation takes two-particle fermion number-conserving operators (i.e., $\hat{c}_i^\dagger \hat{c}_j$ or $\hat{n}_i$) into similar two-particle operators, and a single orbital only has one such operator, $\hat{n}_i$. This applies also to the two-electron terms. If one knows the single particle reduced density matrix (RDM, $\langle \hat{c}_i^\dagger \hat{c}_j \rangle$), one can determine if reducing a particular set of functions to one function is a good approximation: it corresponds to the diagonal block over the sites of the RDM having just one significant eigenvalue. If so, that eigenvector would be the correct linear combination. One may expect that the outer tail regions of a molecule would satisfy this criterion since far from the nuclei, the wavefunction decays in a simple way. This suggests that one should combine sites in the tail regions into tail functions, which could lead to a large reduction in the number of basis functions. If one transforms a set of basis function into a few functions, this breaks the diagonal approximation, but it only generates a limited number of off-diagonal terms, which may be acceptable. Specifically, if basis function i is in the reduced set and p, q, and r are outside the set, terms such as $\hat{c}_i^\dagger \hat{c}_p^\dagger c_q c_r$ with only one (or three) operators in the set are not generated.

A very important question is how to make a diagonal or near-diagonal approximation with a basis composed of an array of gausslets with some additional non-gausslet functions, particularly to represent the sharp behaviour near nuclei. We leave this question for future studies.

VII. DISCUSSION

Application of these ideas to sliced basis DMRG would require little development beyond what is presented here. The initial applications of SBDMRG used a finite difference grid

TABLE II. Coefficients b_j of the Gaussians defining the gausslet $\mathcal{G}_4$ for $j \geq 0$, where $b_{-j} = b_j$.

j	b_j	j	b_j	j	b_j	j	b_j
0	0.606 768 623 902 971 8	12	−0.000 763 458 149 183 9	24	−0.000 000 011 185 763 8	36	−0.000 000 000 000 000 1
1	0.459 576 273 139 799 2	13	0.000 165 471 854 679 3	25	0.000 000 005 757 018 4	37	0.000 000 000 000 000 0
2	−0.016 442 719 020 440 5	14	−0.000 106 297 460 303 6	26	0.000 000 001 754 409 3	38	0.000 000 000 000 000 3
3	−0.140 361 865 534 505 0	15	0.000 148 973 049 197 6	27	−0.000 000 007 839 649 3	39	0.000 000 000 000 000 4
4	−0.040 293 940 178 627 9	16	−0.000 092 552 726 767 9	28	0.000 000 002 920 995 3	40	0.000 000 000 000 000 3
5	0.009 134 592 371 513 9	17	0.000 058 914 812 901 2	29	−0.000 000 001 115 959 4	41	−0.000 000 000 000 000 4
6	0.041 207 471 687 590 8	18	−0.000 034 396 209 858 1	30	0.000 000 000 664 634 6	42	−0.000 000 000 000 000 5
7	−0.026 310 423 100 181 4	19	0.000 013 946 580 494 1	31	−0.000 000 000 108 079 8	43	−0.000 000 000 000 000 4
8	0.012 039 082 210 767 3	20	−0.000 005 590 060 341 2	32	−0.000 000 000 000 000 0	44	−0.000 000 000 000 000 1
9	−0.004 112 077 608 479 4	21	0.000 002 199 375 247 0	33	−0.000 000 000 000 000 0	45	−0.000 000 000 000 000 1
10	0.000 315 581 418 234 8	22	−0.000 000 777 315 756 5	34	−0.000 000 000 000 000 0	46	−0.000 000 000 000 000 1
11	0.000 890 526 535 932 6	23	0.000 000 182 131 145 8	35	0.000 000 000 000 000 0	47	0.000 000 000 000 000 0
12	−0.000 763 458 149 183 9	24	−0.000 000 011 185 763 8	36	−0.000 000 000 000 000 1	48	0.000 000 000 000 000 1

TABLE III. Coefficients of the Gaussians defining the gausslet $\mathcal{G}_6$.

j	b_j	j	b_j	j	b_j	j	b_j
0	0.651 079 912 213 856 5	10	−0.012 999 513 205 108 5	20	0.000 052 757 953 984 0	30	0.000 000 001 223 344 1
1	0.374 890 195 133 727 0	11	0.008 344 462 114 533 6	21	−0.000 028 480 265 623 0	31	0.000 000 006 386 355 5
2	0.093 939 943 721 432 9	12	−0.003 560 204 526 660 4	22	0.000 015 001 501 527 2	32	−0.000 000 003 068 421 5
3	−0.156 900 646 562 756 9	13	0.001 254 495 950 154 9	23	−0.000 006 880 832 116 1	33	0.000 000 000 450 045 7
4	−0.094 815 552 775 120 6	14	0.000 359 462 765 580 7	24	0.000 002 800 455 509 1	34	−0.000 000 000 221 804 0
5	0.023 264 625 608 686 0	15	−0.000 784 842 380 900 6	25	−0.000 001 306 734 674 3	35	0.000 000 000 091 388 2
6	0.021 661 376 830 479 2	16	0.000 574 714 810 259 2	26	0.000 000 716 855 412 3	36	0.000 000 000 010 456 9
7	0.036 180 502 106 294 6	17	−0.000 307 005 329 797 1	27	−0.000 000 391 205 459 9	37	−0.000 000 000 004 399 2
8	−0.031 714 898 150 240 8	18	0.000 149 988 658 848 0	28	0.000 000 161 354 668 6	38	−0.000 000 000 000 000 0
9	0.013 391 581 406 505 9	19	−0.000 089 565 465 873 5	29	−0.000 000 040 165 016 6	39	−0.000 000 000 000 000 0
10	−0.012 999 513 205 108 5	20	0.000 052 757 953 984 0	30	0.000 000 001 223 344 1	40	−0.000 000 000 000 000 0

TABLE IV. Coefficients of the Gaussians defining the gausslet $\mathcal{G}_8$.

j	b_j	j	b_j	j	b_j	j	b_j
0	0.618 848 936 127 006 5	12	−0.004 654 727 525 475 3	24	0.000 028 902 034 112 5	36	0.000 000 006 701 673 8
1	0.382 416 745 427 370 2	13	0.004 010 020 108 354 1	25	−0.000 016 882 968 243 5	37	−0.000 000 003 207 504 6
2	0.109 947 489 746 558 0	14	−0.001 518 470 003 454 9	26	0.000 009 711 908 464 2	38	0.000 000 000 915 363 9
3	−0.147 865 470 727 970 2	15	−0.000 247 352 088 447 7	27	−0.000 005 429 494 846 8	39	−0.000 000 000 150 813 3
4	−0.109 253 317 589 479 7	16	0.000 402 121 244 188 0	28	0.000 002 790 562 289 9	40	0.000 000 000 057 577 2
5	0.000 835 087 680 518 8	17	−0.000 670 985 512 181 6	29	−0.000 001 165 669 266 8	41	−0.000 000 000 014 779 9
6	0.038 346 851 375 262 4	18	0.000 647 992 983 452 6	30	0.000 000 425 130 482 7	42	−0.000 000 000 004 106 9
7	0.044 379 386 734 827 1	19	−0.000 407 227 542 394 3	31	−0.000 000 191 817 897 4	43	0.000 000 000 001 557 2
8	−0.026 422 070 509 827 9	20	0.000 250 705 939 695 2	32	0.000 000 101 185 391 4	44	0.000 000 000 000 000 7
9	0.003 944 539 049 070 3	21	−0.000 149 013 036 761 8	33	−0.000 000 052 108 419 5	45	0.000 000 000 000 000 2
10	−0.018 091 592 177 504 4	22	0.000 085 293 624 090 8	34	0.000 000 026 638 816 7	46	0.000 000 000 000 000 1
11	0.013 034 598 997 590 0	23	−0.000 049 880 431 410 9	35	−0.000 000 012 927 275 4	47	0.000 000 000 000 000 2
12	−0.004 654 727 525 475 3	24	0.000 028 902 034 112 5	36	0.000 000 006 701 673 8	48	0.000 000 000 000 000 0

which was adapted with a low-frequency filtering procedure to increase convergence with the grid spacing. Usually a grid spacing of 0.1 bohr was used for hydrogen chains. The grid plus filtering procedure could be replaced by a gausslet basis, with a diagonal approximation introduced for the two-electron interaction. Probably this would allow substantially higher lattice spacing, and it would smooth the way for including larger Z atoms.

TABLE V. Coefficients of the Gaussians defining the gausslet $\mathcal{G}_{10}$.

j	b_j	j	b_j	j	b_j	j	b_j
0	0.600 628 229 278 303 1	17	−0.000 969 516 111 426 0	34	0.000 000 413 999 191 0	51	−0.000 000 000 000 003 0
1	0.387 090 413 205 924 9	18	0.001 238 162 074 865 4	35	−0.000 000 186 952 757 6	52	−0.000 000 000 000 008 8
2	0.116 743 609 510 183 7	19	−0.000 865 751 227 079 5	36	0.000 000 078 731 044 9	53	0.000 000 000 000 006 3
3	−0.140 114 197 851 207 2	20	0.000 705 059 075 044 2	37	−0.000 000 036 081 218 9	54	−0.000 000 000 000 001 9
4	−0.117 855 298 379 461 4	21	−0.000 532 297 906 670 5	38	0.000 000 016 852 562 8	55	0.000 000 000 000 003 0
5	−0.011 263 261 809 470 0	22	0.000 333 287 449 565 9	39	−0.000 000 008 704 088 3	56	−0.000 000 000 000 001 8
6	0.045 056 014 498 175 7	23	−0.000 217 803 210 413 9	40	0.000 000 004 225 524 2	57	0.000 000 000 000 000 2
7	0.050 213 166 699 230 6	24	0.000 138 960 841 118 4	41	−0.000 000 001 463 924 6	58	−0.000 000 000 000 000 1
8	−0.020 737 279 949 598 2	25	−0.000 084 954 392 328 9	42	0.000 000 000 647 345 1	59	−0.000 000 000 000 000 4
9	−0.003 181 462 446 422 4	26	0.000 053 351 501 075 0	43	−0.000 000 000 373 985 6	60	0.000 000 000 000 000 5
10	−0.021 490 013 694 258 3	27	−0.000 032 797 105 416 6	44	0.000 000 000 141 986 3	61	−0.000 000 000 000 000 4
11	0.013 936 930 862 720 8	28	0.000 019 327 821 407 5	45	−0.000 000 000 070 556 4	62	0.000 000 000 000 000 3
12	−0.002 959 434 007 223 3	29	−0.000 010 867 460 417 1	46	0.000 000 000 041 805 4	63	−0.000 000 000 000 000 3
13	0.005 704 671 223 315 2	30	0.000 006 021 335 304 3	47	−0.000 000 000 009 763 9	64	0.000 000 000 000 000 1
14	−0.002 681 933 418 588 2	31	−0.000 003 314 039 628 2	48	−0.000 000 000 000 875 5	65	−0.000 000 000 000 000 1
15	−0.000 461 190 220 335 7	32	0.000 001 680 135 825 8	49	0.000 000 000 000 477 6	66	0.000 000 000 000 000 0
16	0.000 320 566 229 920 2	33	−0.000 000 824 253 488 7	50	−0.000 000 000 000 006 1	67	−0.000 000 000 000 000 1
17	−0.000 969 516 111 426 0	34	0.000 000 413 999 191 0	51	−0.000 000 000 000 003 0	68	0.000 000 000 000 000 0

TABLE VI. Coefficients of wavelet transform W_{220}, for the scaling function c_i, and the middle and high momentum wavelets, for $i \geq 0$. The scaling function is even, whereas the other two are even or odd about $i = 1.5$.

i	c_i	mid	high
0	0.748 417 015 616 181 5	−0.414 180 060 185 368 8	−0.202 495 761 426 328 7
1	0.427 666 866 066 389 5	0.556 194 593 394 828 8	0.677 157 049 828 566 0
2	0.171 066 746 426 555 8	0.556 194 593 394 828 8	−0.677 157 049 828 566 0
3	−0.085 533 373 213 277 9	−0.414 180 060 185 368 8	0.202 495 761 426 328 7
4	−0.021 383 343 303 319 5	−0.137 565 705 918 330 8	0.016 603 249 484 593 6
5	...	−0.011 863 539 443 011 5	−0.011 863 539 443 011 5
6	...	0.005 931 769 721 505 7	0.005 931 769 721 505 7
7	...	0.001 482 942 430 376 4	0.001 482 942 430 376 4

TABLE VII. Coefficients as above for W_{212}.

i	c_i	mid	high
0	0.696 923 425 058 675 9	−0.410 794 129 495 830 9	−0.291 220 973 626 780 8
1	0.492 799 279 826 744 4	0.556 931 876 836 551 3	0.641 482 866 199 432 6
2	0.102 062 072 615 965 8	0.556 931 876 836 551 3	−0.641 482 866 199 432 6
3	−0.059 786 577 934 525 1	−0.410 794 129 495 830 9	0.291 220 973 626 780 8
4	−0.017 511 083 253 084 4	−0.145 083 226 286 060 4	0.060 532 236 923 179 1
5	...	−0.004 346 019 075 281 8	−0.004 346 019 075 281 8
6	...	0.002 545 839 031 967 9	0.002 545 839 031 967 9
7	...	0.000 745 658 988 654 0	0.000 745 658 988 654 0

More generally, for realistic 3D systems, one can form a 3D gausslet basis by making products of 1D gausslets, namely,

$$\mathcal{G}(x, y, z) = \mathcal{G}(x)\mathcal{G}(y)\mathcal{G}(z), \tag{29}$$

If one is not worried about making diagonal approximations, then one can combine gausslets with 3D Gaussians from a standard Gaussian basis, orthogonalizing them to the gausslets. The fact that gausslets are made of Gaussians would facilitate this. Note that for a uniform grid of gausslets, all the two-electron integrals between gausslets need be done only once and would be applicable to all systems, with a simple rescaling for different lattice spacings.

Diagonal approximation can be extremely effective in speeding up calculations, and so one would want to try to find approaches that allow them. The most straightforward approach would involve using a pseudopotential both of the usual one-electron type and also a two-electron pseudopotential.[17] This one-electron pseudopotential means that a gausslet 3D grid, without supplementary Gaussians, would be adequate, and the two-electron pseudopotential would allow any of our diagonal approximation for the two-electron interaction. The two-electron diagonal approximation would greatly speed up DMRG and also allow other tensor network methods, such as projected entangled pair states (PEPS), to be tried. The diagonal interaction could be compressed for use in DMRG, just as is done in SBDMRG. The diagonal approximation may also speed up other correlation methods. To the best of our knowledge, the combination of both types of pseudopotentials

TABLE VIII. Coefficients as above for W_{432}.

i	c_i	mid	high
0	0.665 451 815 464 682 6	−0.461 771 790 179 972 5	−0.235 987 417 908 452 1
1	0.496 988 909 160 537 8	0.518 654 152 471 527 5	0.658 599 842 666 627 4
2	0.152 241 744 280 667 1	0.518 654 152 471 527 5	−0.658 599 842 666 627 4
3	−0.059 391 997 777 842 4	−0.461 771 790 179 972 5	0.235 987 417 908 452 1
4	−0.067 235 449 653 067 0	−0.107 863 717 540 707 3	0.084 160 709 523 621 2
5	−0.009 391 741 678 102 1	−0.039 714 569 037 649 0	0.020 920 410 174 110 8
6	0.015 735 804 796 996 6	0.055 763 090 522 622 8	−0.052 667 050 723 528 2
7	0.005 202 112 344 308 8	0.037 862 155 679 992 3	−0.016 090 899 407 671 6
8	0.000 132 448 007 801 2	0.001 111 399 595 488 8	0.000 330 246 149 065 8
9	−0.000 394 580 156 682 7	−0.002 273 042 208 353 7	0.000 054 117 364 672 9
10	−0.000 587 753 272 519 8	−0.001 839 730 211 409 6	0.001 626 728 077 374 2
11	...	−0.000 011 228 571 576 3	−0.000 011 228 571 576 3
12	...	0.000 033 451 401 840 3	0.000 033 451 401 840 3
13	...	0.000 049 828 078 196 6	0.000 049 828 078 196 6

has not been used before, but it seems reasonably straightforward. The key question would be: what grid spacing is needed for reasonable accuracy?

For all electron calculations combining gausslets and 3D Gaussians, two-electron diagonal approximations would need to be tested and developed since the extra functions do not fit within the δ-function framework we used to derive diagonal approximations. Nevertheless, one might find approximations with acceptable accuracy. One could also consider whether diagonal approximations could be useful even in a standard Gaussian basis which has been localized by a standard method. Recently, Baker, Burke, and White[22] have proposed "wavelet

TABLE IX. Coefficients as above for W_{652}.

i	c_i	mid	high
0	0.650 187 225 387 146 1	−0.332 669 132 036 125 6	−0.398 552 353 904 044 1
1	0.494 096 412 076 875 0	0.461 713 059 829 481 1	0.529 422 527 269 408 4
2	0.178 098 452 843 265 3	0.461 713 059 829 481 1	−0.529 422 527 269 408 4
3	−0.055 023 498 926 255 2	−0.332 669 132 036 125 6	0.398 552 353 904 044 1
4	−0.090 918 689 274 540 2	−0.351 342 914 816 631 1	−0.110 165 640 731 953 3
5	−0.022 329 265 697 958 5	0.212 384 166 046 800 2	−0.172 290 829 958 789 0
6	0.022 872 288 757 578 2	−0.043 505 928 015 805 6	0.112 083 023 232 846 0
7	0.020 769 210 207 351 2	0.047 174 015 956 452 7	0.064 173 478 040 846 9
8	0.000 187 555 290 049 6	0.047 264 976 548 360 1	−0.047 420 585 131 499 9
9	−0.004 667 396 365 731 4	−0.032 178 987 475 593 6	0.006 069 438 129 874 1
10	−0.002 982 747 265 182 7	−0.014 132 049 212 515 4	−0.004 151 012 510 128 5
11	0.000 012 280 432 572 8	0.003 804 547 272 687 6	−0.006 902 034 108 659 9
12	0.000 495 811 078 186 7	0.000 509 799 631 374 5	0.003 828 899 107 975 7
13	0.000 250 642 807 903 8	0.000 816 864 630 321 5	0.001 133 189 076 076 3
14	0.000 190 453 010 476 3	0.000 705 690 170 792 0	−0.000 075 816 980 457 3
15	−0.000 095 682 642 538 5	−0.000 415 701 480 425 5	−0.000 023 076 185 570 7
16	−0.000 024 035 241 186 9	−0.000 119 787 986 753 2	−0.000 021 161 491 836 7
17	...	−0.000 023 206 664 930 4	−0.000 023 206 664 930 4
18	...	0.000 011 658 912 712 9	0.000 011 658 912 712 9
19	...	0.000 002 928 689 797 8	0.000 002 928 689 797 8

TABLE X. Coefficients as above for W_{872}.

i	c_i	mid	high
0	0.640 511 415 592 180 2	−0.337 513 998 047 977 8	−0.403 994 283 478 911 2
1	0.493 988 321 100 699 6	0.450 385 313 713 616 5	0.523 071 281 551 307 4
2	0.188 346 741 922 076 8	0.450 385 313 713 616 5	−0.523 071 281 551 307 4
3	−0.050 964 250 991 421 7	−0.337 513 998 047 977 8	0.403 994 283 478 911 2
4	−0.100 208 794 736 124 0	−0.350 093 477 310 093 0	−0.103 986 049 752 588 9
5	−0.031 987 682 948 838 9	0.226 203 210 661 894 6	−0.182 345 166 585 476 8
6	0.026 426 779 848 046 6	−0.045 708 153 564 492 5	0.105 900 455 718 599 5
7	0.029 641 613 005 993 1	0.068 489 963 227 058 6	0.077 568 413 207 594 1
8	0.003 227 961 262 516 9	0.034 710 096 135 941 0	−0.041 024 900 027 3447
9	−0.008 518 281 555 442 9	−0.031 387 150 644 330 3	−0.004 951 055 683 587 7
10	−0.006 846 248 914 061 1	−0.021 366 704 546 381 8	−0.012 770 139 721 181 9
11	0.000 109 604 466 819 6	−0.003 462 818 348 006 1	0.000 393 922 871 351 0
12	0.001 721 448 564 128 2	0.006 928 850 578 087 0	0.003 223 712 685 385 4
13	0.001 121 513 967 521 6	0.004 073 210 579 117 4	0.002 217 317 275 796 2
14	0.000 170 488 239 860 8	−0.000 079 454 580 741 3	0.001 675 348 848 330 0
15	−0.000 299 865 208 704 3	−0.000 734 044 690 333 1	−0.001 160 688 462 878 5
16	−0.000 145 285 171 980 7	−0.000 454 787 330 344 4	−0.000 406 067 848 951 7
17	−0.000 092 070 823 211 7	−0.000 247 814 079 586 9	−0.000 089 924 757 066 5
18	0.000 057 776 321 073 1	0.000 185 030 627 530 3	0.000 077 298 755 451 1
19	0.000 017 981 057 761 0	0.000 065 528 792 298 8	0.000 029 752 379 989 9
20	0.000 007 320 286 085 1	0.000 032 701 342 024 5	0.000 007 967 021 505 5
21	−0.000 004 180 178 956 2	−0.000 019 443 298 179 8	−0.000 005 319 002 363 0
22	−0.000 001 193 525 492 2	−0.000 005 771 165 233 6	−0.000 001 738 393 600 2
23	...	−0.000 001 083 243 591 1	−0.000 001 083 243 591 1
24	...	0.000 000 618 575 833 2	0.000 000 618 575 833 2
25	...	0.000 000 176 615 889 8	0.000 000 176 615 889 8

localization" (WL), where an auxiliary wavelet basis is used to localize an existing delocalized basis, at the cost of a modest increase in the number of functions in the basis. One could equally use gausslets to perform wavelet localization. One scheme would be to WL to localize standard Gaussian atom-centered basis functions on each atom. The result would be functions which look like standard Gaussians close to their nucleus but midway to the next atom they would rapidly die off with oscillations to make them orthogonal to all functions on neighboring atoms. This might give rise to a modified diagonal approximation in which V_{ijkl} is nonzero only if i and l are on the same atom and j and k are on the same atom.

In conclusion, we have introduced gausslets, a new type of basis function, which combine the efficiencies of working with Gaussians, but with systematic completeness and orthogonality, while maintaining locality, symmetry, and smoothness. We have introduced diagonal approximations, which are tied to gausslets, which dramatically improve the scaling of electronic structure calculations, making them act more like a grid than a basis. Although the various tests we have performed here are in 1D, these basis functions were developed with 3D applications in mind, and we anticipate rapid development of 3D uses.

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APPENDIX: COEFFICIENTS OF WAVELETS AND GAUSSLETS

Here we give coefficients for the wavelets and gausslets described here. First, we present the coefficients of the gausslets, which are defined by

$$\mathcal{G}(x) = \sum_j b_j \exp[-\frac{1}{2}(3x - j)^2]. \tag{A1}$$

b_j for gausslets with even order 4 through 10 are given in Tables II–V.

In Tables VI–XI, the coefficients defining ternary wavelet transforms are given. These can be used to define sequence of gausslets based on the primary gausslets given in Tables II–V. For example, applying a WT to a primary gausslet gives a

TABLE XI. Coefficients as above for W_{1092}.

i	c_i	mid	high
0	0.633 967 863 084 693 5	−0.271 244 428 864 689 9	−0.437 612 024 071 427 2
1	0.493 732 579 953 340 4	0.403 918 233 936 210 1	0.442 572 391 737 879 4
2	0.194 709 272 052 529 2	0.403 918 233 936 210 1	−0.442 572 391 737 879 4
3	−0.047 539 962 433 007 2	−0.271 244 428 864 689 9	0.437 612 024 071 427 2
4	−0.105 850 410 621 274 8	−0.400 361 290 533 790 4	−0.148 233 958 394 164 7
5	−0.038 867 420 054 301 0	0.283 638 295 204 601 3	−0.225 224 164 221 834 7
6	0.028 004 559 999 195 6	−0.086 940 210 305 369 5	0.141 506 723 188 469 6
7	0.036 026 757 998 886 3	0.080 862 738 894 403 3	0.114 648 710 866 273 1
8	0.006 475 547 896 335 9	0.062 511 314 136 446 9	−0.077 915 968 029 983 9
9	−0.011 460 695 923 810 3	−0.054 298 848 667 428 4	0.006 585 707 069 860 2
10	−0.010 521 136 055 850 7	−0.037 557 877 929 025 6	−0.019 832 224 638 278 8
11	−0.000 402 924 183 783 6	0.006 771 860 788 027 8	−0.008 009 379 762 269 5
12	0.003 301 983 027 767 3	0.007 228 913 625 759 4	0.011 654 189 841 610 0
13	0.002 392 617 491 538 4	0.007 259 011 219 547 3	0.006 706 509 682 697 8
14	0.000 134 199 188 586 9	0.002 731 543 352 820 2	−0.000 248 751 021 745 7
15	−0.000 753 848 805 721 9	−0.003 360 218 601 461 6	−0.001 727 964 658 079 2
16	−0.000 432 959 478 152 9	−0.001 660 005 214 581 3	−0.001 061 464 272 676 8
17	−0.000 147 873 812 113 4	−0.000 202 924 601 199 1	−0.000 777 798 540 500 9
18	0.000 167 773 097 003 9	0.000 466 984 322 661 2	0.000 566 727 556 034 5
19	0.000 070 540 097 154 9	0.000 230 867 212 655 3	0.000 201 869 063 915 1
20	0.000 046 292 473 006 0	0.000 157 423 488 650 6	0.000 067 276 062 439 4
21	−0.000 031 832 101 700 8	−0.000 111 864 960 175 0	−0.000 057 639 826 918 9
22	−0.000 010 270 655 718 8	−0.000 037 774 707 341 0	−0.000 020 860 307 381 0
23	−0.000 005 965 666 564 3	−0.000 019 123 374 266 6	−0.000 015 488 193 549 7
24	0.000 003 464 863 024 5	0.000 012 626 068 515 9	0.000 008 788 696 294 0
25	0.000 000 968 002 586 7	0.000 003 883 946 958 3	0.000 002 486 917 439 7
26	0.000 000 510 369 555 8	0.000 002 510 824 776 9	0.000 000 894 882 144 9
27	−0.000 000 238 670 284 9	−0.000 001 280 771 765 3	−0.000 000 525 088 978 2
28	−0.000 000 055 806 135 2	−0.000 000 324 397 749 1	−0.000 000 147 703 210 7
29	...	−0.000 000 080 596 018 2	−0.000 000 080 596 018 2
30	...	0.000 000 037 690 090 3	0.000 000 037 690 090 3
31	...	0.000 000 008 812 736 3	0.000 000 008 812 736 3

gausslet with Gaussian spacing of 1/9. Combining Eqs. (A1) and (10), we have

$$\mathcal{G}'(x) = \sum_l \exp[-\frac{1}{2}(9x - l)^2] \sum_k c_k b_{l-3k}, \qquad \text{(A2)}$$

where b_j come from the primary gausslet and c_k are the coefficients associated with the additional WT. To define wavelet-like functions coming from this transformation, we use the same formula, but with the mid or high wavelet coefficients replacing c_k. One can repeat these WTs to construct a multi-level wavelet-like basis, where the functions at each scale, in addition to different scalings, are slightly different. All the basis functions in the multi-level basis would be written in terms of a single grid of Gaussians, which would have a spacing a factor of three smaller than the finest level of wavelet-like functions.

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